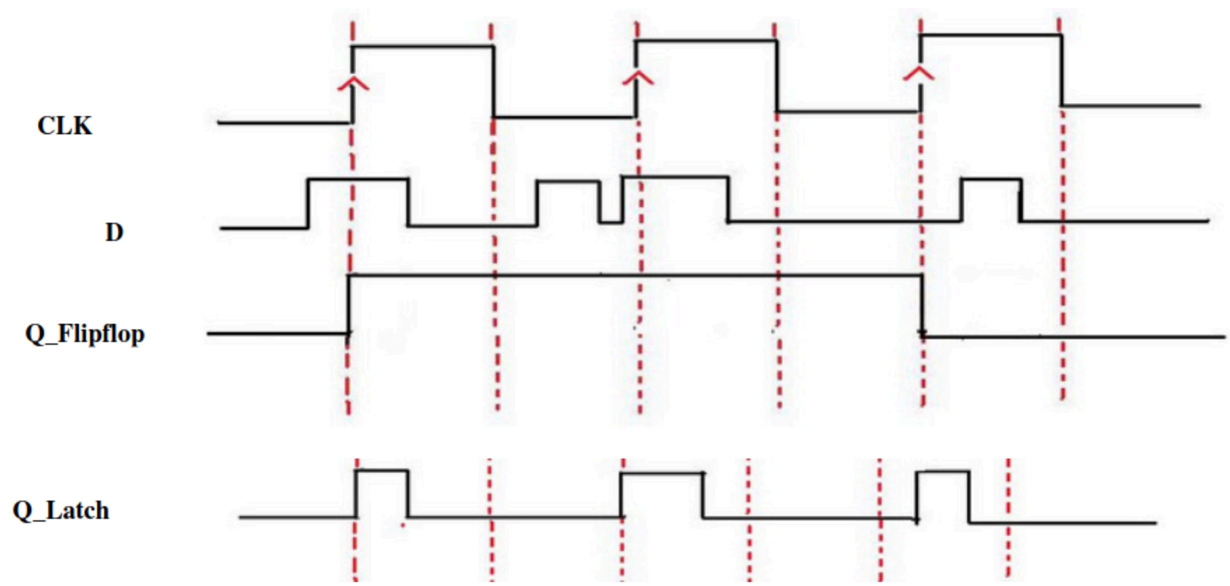
Circuit Simulation Lab:12
Understanding the difference of Latches and Flipflop



Design a verilog program such a way that it has ports clk, D, Q_Latch, Q_Flipflop outputs and capture the waveforms

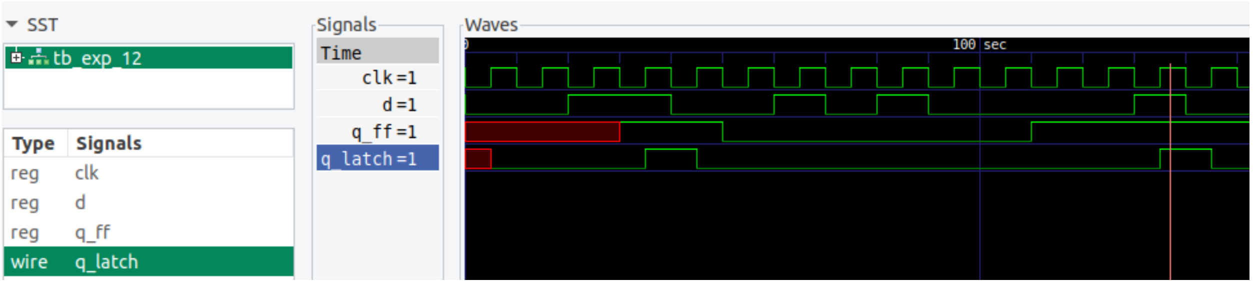
CODE:

```
module exp_12(
input clk , d,q_ff,
output reg q_latch );

always @(posedge clk) begin
if (d == 1 && q_ff == 1 )
q_latch <= 1 ;
else
q_latch <= 0 ;
end
endmodule
```

TESTBENCH

```
module tb_exp_12;
wire q_latch;
reg clk , d, q_ff;
```

```
exp_12 e1 (
.q_latch(q_latch),.clk(clk),.d(d),.q_ff(q_ff));

always #5 clk = ~clk;

initial begin
$dumpfile ("exp12.vcd");
$dumpvars (0,tb_exp_12);

d   = 0;
clk = 0;
#20  d = 1;
#10; q_ff= 1 ;

#10  d = 0;
#10; q_ff= 0 ;

#10  d = 1;
#10  d = 0;


#10  d = 1;
#10  d = 0;
#20; q_ff= 1 ;

#10  d = 0;



#10  d = 1;
#10  d = 0;



#20;
$finish;

end
endmodule
```


SST
tb_exp_12
Type Signals
reg clk
reg d
reg q_ff
wire q_latch
Signals
Time
clk=1
d=1
q_ff=1
q_latch=1
Waves
100 sec